

KL Deemed to be University Aziz Nagar Campus

Department of ECE

1st Review

MD Sameer Ahmed - 2310040004

Sayooj S - 2310040095

Supervised by:

Dr. Ngangbam Phalguni Singh, Assoc. Prof./ECE

Introduction :

- Edge AI applications require efficient processors for fast, low-power computation on embedded devices.
- RISC-V enables customizable processor architectures through flexible instruction set extensions and open-source designs.
- This project develops a configurable FPGA-based AI accelerator integrated with a PicoRV32 RISC-V processor.

Problem Statement :

- Embedded RISC-V processors execute AI workloads with high latency and limited computational efficiency.
- Software execution of matrix operations increases execution cycles, power consumption, and processing time.
- Configurable hardware acceleration is required for efficient AI processing in embedded systems.

Objectives :

- Design a configurable MAC-based accelerator for an embedded PicoRV32 RISC-V processor.
- Integrate custom RISC-V instructions to accelerate AI-oriented computational kernels efficiently.
- Evaluate performance, area, and power across multiple FPGA accelerator configurations.

Scope of the Project :

- Develop a configurable AI accelerator integrated with the PicoRV32 RISC-V processor on FPGA.
- Implement and verify custom instructions using SystemVerilog through simulation and hardware testing.
- Evaluate performance, resource utilization, and power across multiple accelerator configurations.

Literature Review :

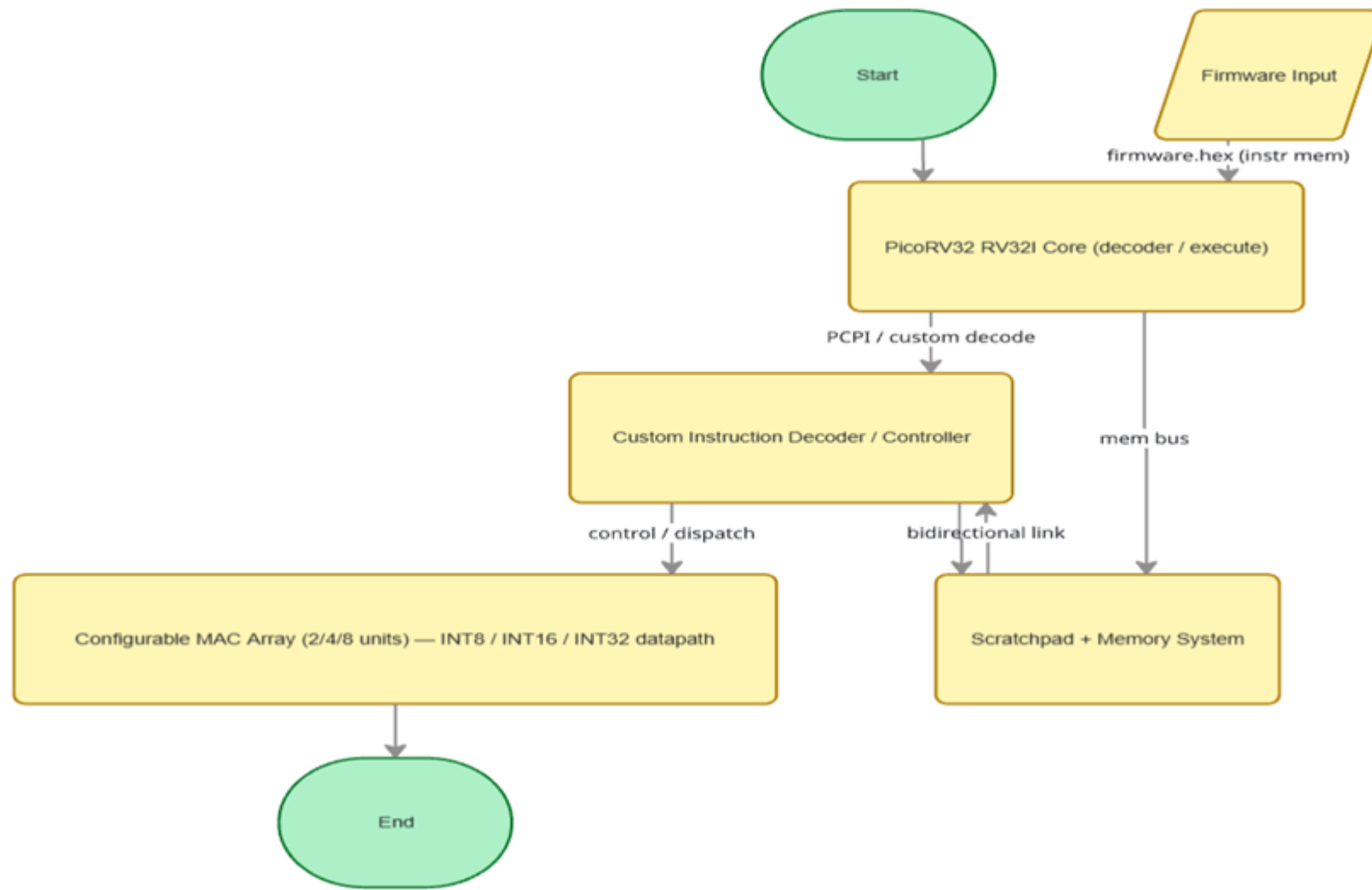
- Studied existing RISC-V processors and custom instruction extension techniques.
- Reviewed hardware accelerator architectures for AI and embedded computing applications.

Research Gap :

- Existing AI accelerators lack flexibility for different embedded application requirements.
- Efficient custom instruction integration for embedded AI applications requires further investigation.

Methodology :

- Initialize PicoRV32 processor with firmware and custom instruction support.
- Decode instructions and dispatch AI operations to the configurable MAC accelerator.
- Verify functionality and evaluate performance using simulation and FPGA implementation.



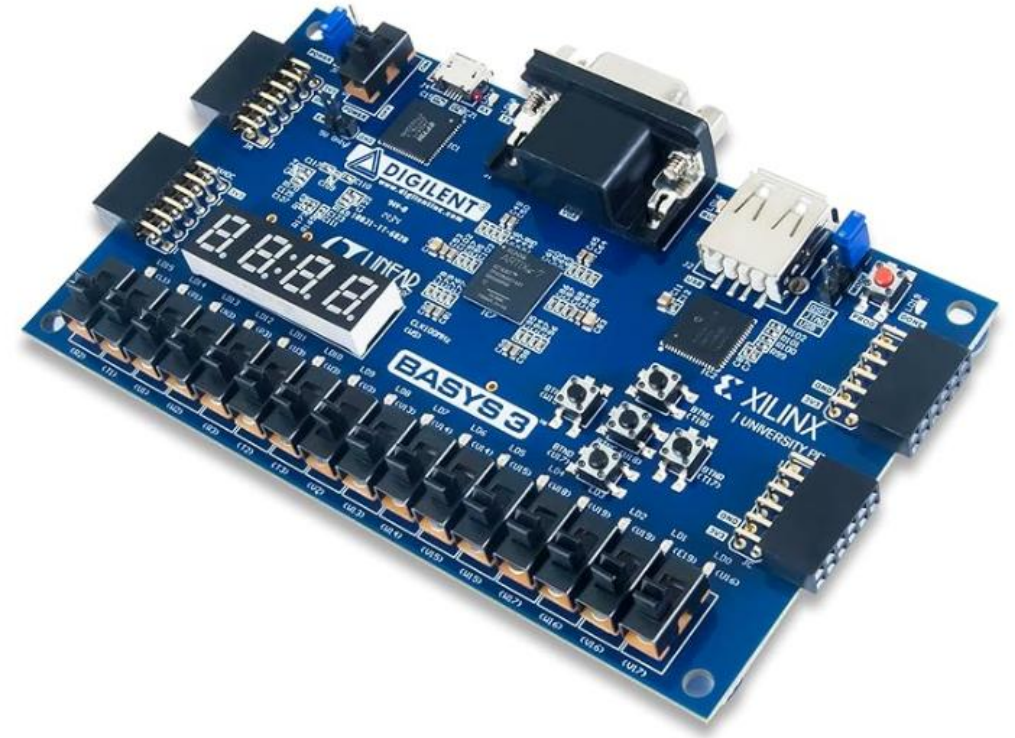
Proposed architecture — configurable AI acceleration extension for an embedded RISC-V processor on FPGA.

Expected Outcomes :

- Develop a configurable AI accelerator integrated with the PicoRV32 RISC-V processor.
- Demonstrate improved performance using custom instructions and hardware acceleration on FPGA.
- Analyze speed, area, and power trade-offs across multiple accelerator configurations.

Financial Arrangements :

- Basys 3 FPGA board allocated for hardware implementation and prototyping.
- Required accessories include USB cable, power supply, and jumper wires.
- Estimated project budget is approximately ₹27,578



Duration :

SL.NO.	TASK NAME	2026					
		JUL	AUG	SEP	OCT	NOV	DEC
1	Literature review	✓					
2	Data collection & system analysis		✓	✓			
3	System Design and Development			✓	✓		
4	Prototype testing & installation				✓	✓	
5	Writing report					✓	✓
6	Submission						✓

Expected time line schedule

References :

- E. Rezunov, N. Zurstraßen, L. M. Reimann, and R. Leupers, "Automatic Microarchitecture-Aware Custom Instruction Design for RISC-V Processors," in Proc. IEEE/ACM Int. Conf. Computer-Aided Design (ICCAD), Munich, Germany, 2025, pp. 1–9. doi: 10.1109/ICCAD66269.2025.11240781.
- K. S. Gorde, S. M. Sonavane, S. Hutke, and A. Hutke, "Analog artificial intelligence hardware for neural networks: Design trends and considerations," Bulletin of Electrical Engineering and Informatics, vol. 14, no. 6, pp. 4399–4410, Dec. 2025. doi: 10.11591/eei.v14i6.10842.
- M. Damian, J. Oppermann, C. Spang, and A. Koch, "SCALE-V: An Open-Source SCALable Interface for ISA Extensions for RISC-V Processors," in Proc. 59th ACM/IEEE Design Automation Conf. (DAC), San Francisco, CA, USA, 2022, pp. 1–6. doi: 10.1145/3489517.3530432.

